Beyond Automation: AI-Driven Project Management with OpenAI and Prompt Engineering

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***Abstract*—Managing project management tools is manual and lacks specific integrations of the Large Language Model making the entire process of retrieval and feeding useful information time-consuming, prone to errors and inefficient. This paper is an innovative approach to enhance project management and collaboration by integrating OpenAI's AI with web-based tools, utilizing Prompt Engineering. Team members interact through simple prompts, facilitating tasks such as assignment, tracking, and resource allocation to querying useful information from context in natural language using specific prompts. The web-based system ensures accessibility and real-time updates, incorporating Next JS for versatility and integration with different project management methodologies.**

Keywords—OpenAI, prompt engineering, web-based, Next JS, Project management, AI, teamwork.

I. Introduction

The "AI and Web-Based Teamwork Project Management Tool" leverages OpenAI and Prompt Engineering to revolutionize project management [1], [2]. Users interact effortlessly through prompts, spanning task assignments to milestone monitoring [3], [4]. The web-based tool, powered by ReactJS and Next JS, ensures seamless accessibility across devices. Its adaptability, integrating with diverse project management methodologies, makes it a versatile asset. The comprehensive tech stack includes Tailwind CSS, REST APIs, OpenAI, Langchain, and Context-based search, providing a user-friendly and responsive experience.

The conventional project management tools relied heavily on the user's instincts, like Trello and GitHub projects. Where the user did most of the heavy lifting. The crux of project management lies in efficient and quick information extraction to make better decisions and elevate the product market fit for any organization. Using the existing techniques, we felt that the crucial work of decision-making wasn't automated. Part of that had to do with the fact that for effective information retrieval, the system had to be aware of the entire context and make crucial decisions based on this context. The major hurdle was analyzing the context written completely in Natural Language. When it came to analyzing natural language, there came many hurdles. Like, just to identify the overall sentiment of a statement in natural language, we cannot just rely on the words used, as there can be sarcasm involved. The word bank might never be up to date, as there can be new lingo popping up now and then. Hence, for third-party companies, it became extremely hard to provide its users with the feature of context-based retrieval.

Hence, we concluded to create an entire system, to fill in this gap in the space of project management using the prime technology of context-based retrieval, context-aware AI and making complex decisions using artificial intelligence. Our aim was not to liberate the user from the process of decision-making entirely, but we meant to provide the user with some of the best options to make the decisions from, thereby leading to a decreased time of action in the entire workflow.

Now although, at least at the time of research, both closed as well as open-sourced Generative AI platforms had a common pitfall, that being Hallucination. The research paper discusses how exactly we used context-based information retrieval paradigm coupled with the concepts of mathematical vectorization of data to reduce the latency as well as increase the accuracy of the responses produced by pre-existent Generative AI models [5]. The latter part of the paper discusses maintaining a long-term memory of both the context and the queries of the users [6], [7]. Finally, benchmarks with traditional Gen AI platforms alongside the responses given by our fine- tuned model is elucidated to emphasize on our findings [8], [9], [10].

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II. Objective

The underlying project for this research was to aid the project managers within organizations to provide them with a helpful generative AI and Chat based application in which the non-technical team leads can ask queries regarding the progress of the project, various lawsuits, various acts, information about numerous accessible projects within the organization using the power of Natural Language. Along with this, we also aim to provide them with a web-based GUI to aid them in scheduling various projects.

However, the true objective of the research paper involves using fine-tuned versions of Generative AI to build solutions for context-based retrieval which eliminates the ever-longing problem of Hallucination, also, the ideology to develop a long- term memory by the Gen AI model using underlying RAG systems was developed. The research was also done on the ability of these Gen AI platforms to aid the user with making complex decisions by giving them varied sets of possible options to make an accurate decision.

Each part of the paper emphasizes the findings concerning the problems mentioned above, also, they lay the foundations for performing thorough benchmarks with the pre-existent and comparable solutions. (Some of them being GPT models and Gemini models).

III. Scope

The project addresses the widespread demand for advanced project management tools in diverse industries. In a globalized and tech-driven landscape, the need for innovative AI-driven solutions is critical. This project offers a versatile platform applicable to sectors like IT, construction, marketing, etc., ensuring adaptability across various project management methodologies and providing a user-friendly solution to modern challenges.

The conclusions derived from the paper can be taken as a firm testament to the advancements in Generative AI to understand and comprehend Natural Language with an effort to make crucial decisions with good accuracy.

IV. Proposed System

In the proposed system, we decided to create a web-based application using React JS and backend technologies like Node JS and Django along with a No-SQL data store, specifically Mongo DB with an integration to Open AI's GPT-3.5 model and an unstructured PDF which will be given as a context to the GPT model.

The Gen AI pipeline is proposed to consist of any Gen AI model (however the research was done on GPT-3.5 Turbo, which at the time of testing, was state of the art). The entire point was to make the platform model agnostic, meaning, the choice of choosing different models does not affect the implementation workflow of automation, although, due to different models being differently capable, they might endure a different scale of accuracy.

The context that was fed to Gen AI was done using a vectorized representation of the data, therefore, the context tokens were extremely to the point thereby leading to fewer tokens taken by the context in general.

The process involved taking the user's query and vectorizing it along with the context, then proceeding with finding a similarity metric of the query with the entire context, and then taking just the final results of the previous step to feed it into the GPT transformer as an input to soak up the context. That way, Gen AI's limitation of accepting a limited number of tokens was reduced to a very large extent.

A corresponding pipeline for including RAG to embed long-term memory was also taken into account. The initial approach was to store the previous queries made by the user within a SQL database and fetch the user's queries and passing them each time along with the context as input to the GPT transformer. The approach worked well in the end, giving ~100% accuracy at the final retrievals as it was context-aware of both the user's query as well as the context.

However, we felt a computational inefficiency in doing so, firstly, the database operations cost time and CPU utilization to execute, and secondly, the input token size increased by multiple folds, so to further optimize this, we decided to incrementally embed the vectorized data of user's queries directly to the earlier existent vectorized context.

Advantages:

- Enhanced Task Management and Planning using the GPT model.
- Automation of manual tasks to increase productivity.
- Interaction with the data in Natural Language.

V. System Design

The current implementation involves implementing two personas, the user as well as the co-pilot. Co-pilot is an LLM bridge between the GPT-3.5 Turbo model as well as the User. The user can ask the co-pilot to perform Create, Read, Update and Delete operations on day-to-day task management tasks thereby automating it to a large extent.

The co-pilot is also capable of generating documentation for various types of work as well as it is capable of interacting with the user by keeping a context of data created by the user. The co-pilot also initiates a similarity search to train on the provided context and answers the user's query with pieces of text from the context.

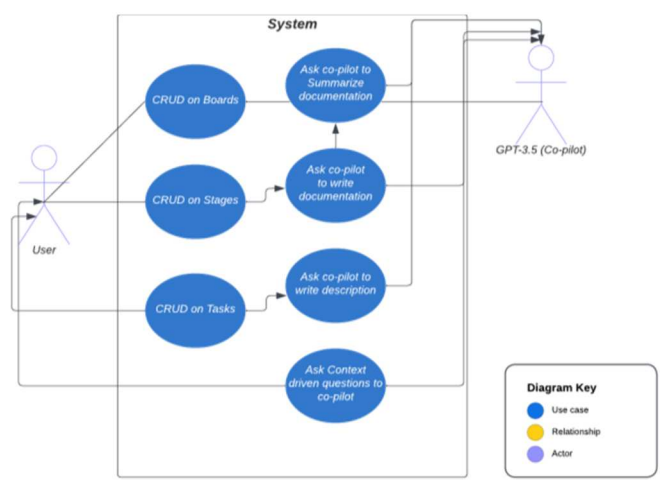

Fig. 1. User and Model interaction

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VI. System Architecture

The client consists of two personas, namely, User and Admin, which communicate with the Server via HTTP protocols, the server consists of a connection to a database server and OpenAI LLM.

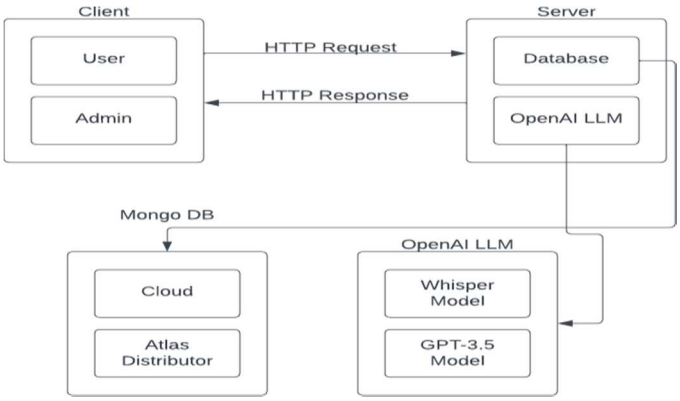

Fig. 2. System Architecture

VII. Implementation And Evaluation Setup

The server-side scripts are written in Python 3.9, and the framework used for building the REST API remains Django. The preliminary focus of the project corresponds to context-based information retrieval to let the user retrieve information from a large corpus of pre-defined text as well as generation of documentation based on prompts provided by the user. The disadvantage of the traditional approach is hallucination, i.e. delivering multiple redundant copies of retrieved information with minor differences leading to inaccurate information retrieval. To solve this, we've used Open AI's text-embedding model, text- embedding-ada-002 to embed the text in a 1536- dimensional vector space. Here, a vector is an array of a 4-byte floating point number, therefore, 1536-dimensional vector space refers to an array of 1536 floating point numbers of 4 bytes. As illustrated below, some sets of tokens are numerically encoded into a vector space spanning 1536 dimensions which represents that particular sequence of tokens.

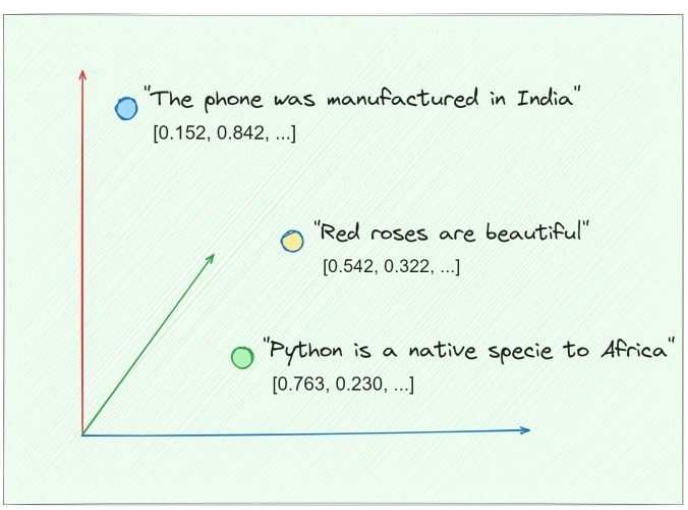

Fig. 3. Graphical representation of embedding of textual data by text-embedding-ada-002 model

The sequence of tokens which are similar to each other will be placed close to each other. Therefore, the lesser the distance between two vector spaces, the more similar they are deemed to be.

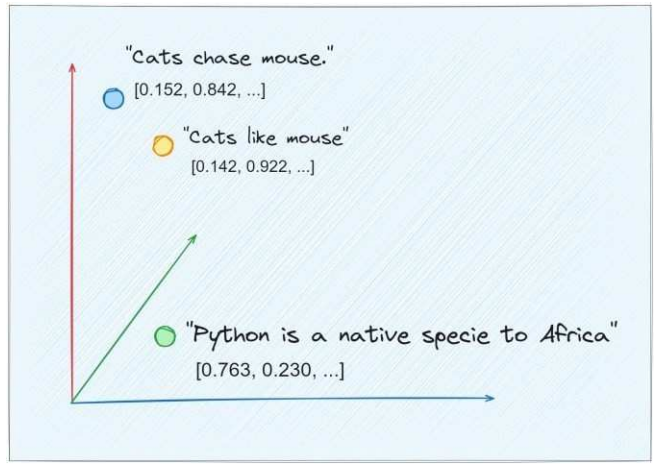

Fig. 4. Graphical representation of embedding of textual data by text-embedding-ada-002 model

To measure this similarity numerically, there are a plethora of similarity metrics. According to Pinecone, a well-established corporation in the domain of vector databases, the majorly used similarity metrics are depicted below.

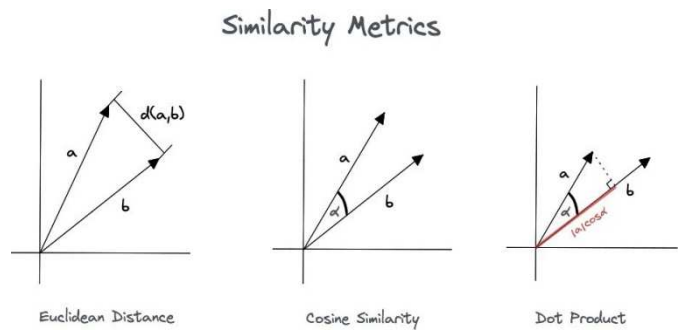

Fig. 5. Graphical representation for some of the available options for similarity metrics

Out of all the similarity metrics available, cosine similarity metric is one of the most widely used similarity metric because of its efficiency in handling sparse data, such as **word documents**. Hence, we found cosine similarity to be perfect for our use case.

The following is the algorithm we used to process the context vectors as well as the user's queries to create a long-term memory and perform similarity checks for performance efficiency.

Step 1: We vectorized the context in 1536 dimensions using the OpenAI's Ada-002 model and stored it inside a vector database.

Step 2: For each query asked by the user, we vectorized the query and performed a similarity check of the query against the already vectorized context. This gave us the idea of the user's requirement from the context, hence the resulting final context coming from the similarity scores ranked in descending order, we chose the one with the highest similarity score assuming that the result with the highest similarity score resembles the user's query more. This was achieved using Langchain's similarity search abstraction. We formalized using cosine similarity for this purpose.

Step 3: We added the vectorized question to the vectorized context of the user thereby storing the past questions within the

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vector databases for long-term memory effect.

Step 4: We fed the result from the similarity search to the GPT Transformer to retrieve GPT's response for the user's query.

Step 5: It was difficult to make the GPT model always answer in JSON format, therefore, we provided a system-level prompt to the GPT model which stated to mandatorily respond in JSON format. A system prompt is different from a normal user or assistant prompt, it lays the guidelines for the Large Language Model to adhere to, which in our case was to generate the results in JavaScript Object Notation (JSON) format. It was therefore guaranteed that all the responses the LLM model made were received in JSON.

All the procedures mentioned above are model agnostic, meaning, they can theoretically be replicated to any model, but for testing and benchmarking, we've usedGPT-3.5 Turbo.

To provide a little insight into maintaining maximum performance in information retrieval, we used LlamaIndex's auto-indexing feature which stores the vectorized embeddings inside a vector database like Pinecone or pgVector by auto-indexing the content for efficient retrieval, similar to the indexing feature often used in SQL databases. This performance technique gave us a massive boost inefficiently retrieving the stored records by the process of context-based retrieval.

Considering future upgradability in mind, Llama Index supports multiple data connectors which would help the user to hook up not only just the structured data coming from SQL tables and other .csv files but rather, expands the horizon by also supporting unstructured as well as semi-structured data (JSON data) coming from APIs, like Notion and Slack. Support for unstructured data enables the user to upload files of different formats, including, but not limited to, PDF files, word files, PowerPoint files, and Excel files as well as web scraped results of a web page expanding the domain of contexts altogether. This further enhances the ease of use of the user with our systems.

The following is the mathematical representation for calculating the cosine similarity.

$$similarity = cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}} \quad (1)$$

A – Vector A

B – Vector B

|| A || – Magnitude of Vector A

|| B || – Magnitude of Vector B

θ – Angle between vectors A and B

Here, Vector A represents the context in vectorized format and vector B represents the user's query in vectorized format, both are represented in 1536 dimensions. Finding cosine similarity then involves finding the dot product and dividing them by the product of their magnitudes.

To summarize, the context i.e. the textual data is first converted into a vector of 1536 dimensions followed by the user's query. A similarity search takes place to determine a similar piece of content in the corpus which is then processed by open AI's GPT-3.5 Turbo and returned to the user in natural language thereby reducing hallucination.

The next segment of our implementation involves prompt engineering the GPT-3.5 Turbo model to produce the desired documentation for the query requested by the user. The prompt that was used for testing out the application was as follows:

"***You are a document generator, generate documentation for the query provided by the user. Make sure the documentation entails all the required requests and remember to generate a comprehensive list of documentation.***

Strictly remember to output the documentation in JSON format. Refer to the following example to generate the documentation: Example:

{

Title: "Heading for the documentation",

Description: "Description for the documentation",

Tags: "Tags corresponding to the documentation",

Methods: [

{

"GET": "Route: /get &^%$ description for the route"

}, {}…

]

}"

Here, the term &^%$ is a separator that separates the route information as well as information about the description of the route.

The requirement involved getting an output in JSON format from the GPT-3.5 Turbo model, hence the line "Strictly remember to output the documentation in JSON format" was provided. However, it was observed that the prompt was found to be almost 90% accurate. To tackle this issue and to increase the efficiency and guarantee of a JSON structured response to be generated, it was later decided to also provide an example statement for the model to learn to respond only in the specified format thereby increasing the accuracy to ~100%.

Illustrative Integration Scenarios:

- **Slack Integration**: Enabling instantaneous communication and collaboration, our tool integrates seamlessly with Slack, a widely adopted team messaging platform. Users can effortlessly receive notifications, updates, and task assignments directly within their Slack channels, thereby streamlining communication channels and enhancing responsiveness.
- **Jira Integration**: Simplifying project tracking and issue management, our solution seamlessly integrates with Jira, a premier agile project management software. This integration ensures seamless synchronization of project tasks, milestones, and user stories between the two platforms, empowering teams to maintain organizational cohesion and focus on deliverables.
- **Microsoft Teams Integration**: Elevating team

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collaboration and document sharing, our tool integrates seamlessly with Microsoft Teams, a unified communication and collaboration platform. With this integration, users gain access to project documents, can initiate video meetings, and collaborate on tasks directly within the Teams interface, fostering a cohesive and productive work environment.

***Advantages of Integration*:**

- **Streamlined Workflows**: Our integration capabilities eradicate manual data entry and streamline project workflows by seamlessly integrating with existing software and platforms within your organization.
- **Enhanced Visibility**: Centralizing data from disparate systems within our AI-driven project management tool provides users with real-time insights into project progress and status, thereby enhancing visibility and transparency.
- **Improved Efficiency**: By consolidating all relevant project information and tools within a single interface, our solution reduces context switching and improves overall efficiency.
- **Enhanced Collaboration**: Our integrations facilitate seamless information sharing and interaction among team members across integrated platforms, fostering a culture of collaboration and teamwork.

In addition to seamless integration with third-party software, our AI-driven project management tool in future can offer advanced task automation capabilities to streamline workflows and boost productivity. Leveraging machine learning algorithms and natural language processing (NLP), our platform automates repetitive tasks, allowing teams to focus on high-value activities and strategic initiatives.

Key Features of Advanced Task Automation:

- **Smart Task Assignment**: Our platform analyzes project requirements, team availability, and skill sets to intelligently assign tasks to the most suitable team members. By considering factors such as workload balance and resource availability, our system ensures efficient task distribution and optimized team performance.
- **Automated Progress Tracking**: With real-time progress tracking and milestone monitoring, our platform automatically updates task statuses and alerts stakeholders of any delays or bottlenecks. By providing visibility into project timelines and dependencies, teams can proactively address issues and keep projects on track.
- **Predictive Resource Allocation**: Using historical project data and predictive analytics, our platform forecasts resource needs and recommends optimal allocation strategies. By anticipating resource shortages or overages, teams can adjust staffing levels and allocate resources more effectively, minimizing project risks and maximizing efficiency.
- **Intelligent Decision Support**: Our platform provides decision support tools powered by AI and machine learning algorithms, helping project managers make informed decisions quickly and confidently. By analyzing data from multiple sources and identifying patterns or trends, our system enables stakeholders to anticipate challenges, mitigate risks, and seize opportunities.
- **Context-Aware Collaboration**: Through advanced NLP capabilities, our platform facilitates context-aware collaboration among team members. By understanding the context of conversations and documents, our system suggests relevant information, prompts, or actions, enhancing communication and knowledge sharing within the team.

By harnessing the power of advanced task automation, our AI-driven project management tool empowers teams to work smarter, not harder, and achieve their project goals with greater efficiency and agility. The objective of the evaluation of this implementation is to Assess the performance and effectiveness of the project management tool in enhancing task management, automating manual tasks, and providing natural language interaction.

The metrics considered in the evaluation process are:

- **Task completion time:** Measure the time taken to complete tasks using the tool compared to traditional methods.
- **Ease of Use:** Measure the intuitiveness of the user interface and navigation flow.
- **Adaptability to Different Workflows:** Assess the flexibility of the tool to accommodate different project management methodologies and workflows, such as Agile, Waterfall, and Kanban.
- **Efficiency:** Measure the efficiency of task execution and resource utilization within the tool, including the time and effort required to complete common project management activities.

The evaluation environment for the AI and web-based project management tool will involve stakeholders from various domains including project managers, team members etc. Performance logs will track task completion times and system performance. Evaluation metrics will include earlier discussed metrics and more.

While exact quantitative measurements were not possible for all aspects of the evaluation, qualitative feedback from users provided valuable insights into their experiences. Here are the qualitative results that we were able to obtain:

- **Ease of Use**: Users found the tool to be intuitive and easy to navigate, with a clean and user-friendly interface.
- **Efficiency**: Users reported significant time savings and increased productivity when using the tool compared to traditional methods.
- **Effectiveness**: Users expressed satisfaction with the tool's effectiveness in facilitating communication and collaboration among team members.
- **Task Completion Time**: Users reported a noticeable reduction in the time required to complete tasks using

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the AI and web-based project management tool compared to traditional methods.

- **Adaptability:** Users praised the tool's adaptability to different project management methodologies

Overall, users were highly satisfied with the tool's usability, efficiency, and effectiveness. They appreciated its versatility, adaptability to different workflows, and integration capabilities.

Therefore, we concluded that to build an effective prompt, it is necessary to provide strict constraints to the model as well as an example scenario for it to learn and respond as we expect it to. This becomes true especially when there are strict constraints in the generated output or if the generated output is strictly formatted in some pre-defined format. (e.g. JSON, YML, XML, etc.)

VIII. Results and Discussion

We tested the final system for long-term memory, context-based retrieval and critical decision-making on the GPT model which was trained on vectorized context beforehand.

When we asked the GPT system which wasn't trained in a specific context to specify the amount of protein intake, we observed hallucinations right from the very beginning of the testing procedures.

For the prompt, "Tell me the amount of protein intake for an Indian of age 21 with 5.6 ft height, 56 kg weight and moderate exercising habits.", the model responded with 48- 60, 98 and 127 grams of protein each day for three different test cases, making the model a poor choice for answering questions like these.

However, later, upon biological research and official data, we curated a protein intake chart that takes in the user's gender, age, and exercise requirements ranging from low, moderate, intense and extreme coupled with his/ her height and weight. The final results were stored in a table where each cell had a specific range of protein intake that it could have. For example, a person with a body weight of 40-60 kilograms, and a height of 5'7'' should entail a protein ranging from 90g – 135g. This context was vectorized and stored in the vector database.

For each subsequent test on the GPT model, the GPT model stopped hallucinating instantly and always gave consistent outputs in sync with the vectorized context therefore improving the entire model by a huge fraction. It was found that the model performed with an accuracy of ~100% with the context inscribed in the query.

The next step involved testing the long-term memory of the conversation, we did this by chaining two prompts where the LLM response for the first prompt was used again on the subsequent prompt to test the long-term memory of the system. And since this was all vector-embedded, out of 100 tests performed on the sample, with a total of 200 prompts, it was found that, for all the 100 test cases, the previous memory of the system was preserved, therefore confirming our prior long-term memory hypothesis. Hence, the accuracy for this was also determined to be ~100%.

This study evaluated a new large language model (LLM), Gemini-1.0, for its ability to predict stock trends. The LLM received historical stock data visualized as candlestick charts and analyzed them for trends. While the model achieved 71% accuracy in predicting trends, it opted not to give financial advice in 19% of cases. This suggests LLMs may be unsuitable for final investment decisions but could provide users with a range of possibilities. The accuracy, though above average, is variable. However, by providing more context and prompting the LLM to generate multiple options, the accuracy for each option could significantly improve, allowing users to make better-informed decisions.

IX. Conclusion

To conclude, Generative AI forms a solid foundation for AI- driven project management. These models can be fine-tuned with regards to a particular context from various third-party integrations as well as textual data and can be coupled with RAG solutions and multi-modality features to make an effective tool in the space of project management. Its ability to make hard decisions effectively can guide the project management in right directions. Also, a prompt library of available prompts can further increase the productivity for daily project management. These solutions coupled together forms a good AI solution which reduces the time to action in any project by a huge margin.

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